

Ola Customer Retention Analysis

Professional Business Intelligence Report

Project Title: Ola Customer Retention Analysis

Domain: Ride-Handling / Mobility Analytics

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Report Type: Business Intelligence & Customer Retention Analysis Report

Tools Used:

- SQL
- Python (Pandas)
- Power BI
- Microsoft Excel (Data Validation)

Dataset Size:

- 150,000 ride booking records
- 21 business attributes
- 148,788 unique customers

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1. Introduction

The ride-sharing industry relies heavily on customer loyalty. Every completed ride contributes to long-term retention, while factors such as frequent cancellations, long wait times, and poor driver experiences can lead to churn. This report evaluates these operational factors to enhance customer retention strategies.

Customers may stop using the platform due to:

- Frequent ride cancellations
- Long waiting time
- Poor driver experience
- High fares
- Unsatisfactory customer service

By examining historical booking datasets, this comprehensive analysis evaluates operational drivers to deliver actionable insights for bolstering customer loyalty.

2. Business Problem

Ola faces customer churn due to operational inefficiencies, including ride cancellations, delayed pickups, and inconsistent service quality. The challenge lies in identifying why customers leave, what payment methods they prefer, and how ratings impact their likelihood to return.

The challenge is to identify:

- Why customers leave
- What impacts ride completion
- Which payment methods customers prefer
- How ratings affect retention
- Which operational issues reduce customer satisfaction

3. Project Objectives

- Analyze booking trends.
- Measure ride completion rates.
- Identify cancellation patterns
- Study customer and driver ratings
- Evaluate revenue generation
- Analyze payment preferences
- Improve customer retention
- Provide business recommendations

4. Dataset Overview

The analysis is based on a robust dataset representing significant operational volume.

Attribute	Value
Total Records	150,000
Total Columns	21
Unique Customers	148,788
Total Revenue	₹51,846,183
Average Booking Value	₹508.30
Average Customer Rating	4.40 / 5

5. Dataset Description

The dataset includes operational ride-level information and major fields such as:

- Booking ID
- Booking Status
- Customer ID

- Vehicle Type
- Pickup Location
- Drop Location
- Booking Value
- Ride Distance
- Payment Method
- Driver Rating
- Customer Rating
- Cancellation Reasons
- Incomplete Ride Reasons
- Vehicle Arrival Time
- Customer Arrival Time

6. Data Cleaning & Preprocessing

To ensure data integrity, the following steps were performed:

- Removed duplicate records
- Checked missing values
- Standardized booking status
- Converted Date column into Date datatype
- Converted numeric columns
- Validated ratings
- Handled null values
- Verified payment methods
- Standardized cancellation reasons

7. Technology Stack

- **SQL:** Used for data extraction, aggregations, and KPI calculations.
- **Python (Pandas):** Utilized for deep data cleaning and exploratory data analysis.
- **Power BI:** Employed for dashboard development and interactive visualizations.

8. Data Model

The Power BI dashboard follows a star-schema-inspired approach with booking records as the primary fact table. Measures were created using DAX to calculate KPIs such as revenue, ride completion rate, cancellation rate, average booking value, customer ratings, and monthly trends.

Interactive slicers allow filtering by:

- Vehicle Type
- Booking Status
- Payment Method
- Date
- Pickup Location

9. KPI Analysis

- **Total Bookings:** Reflects overall customer demand.
- **Completed Rides:** Approximately 93,000 successful rides measured.
- **Cancellation Rate:** A key metric for retention; lower rates indicate better service.
- **Total Revenue:** ₹51.85 Million generated.

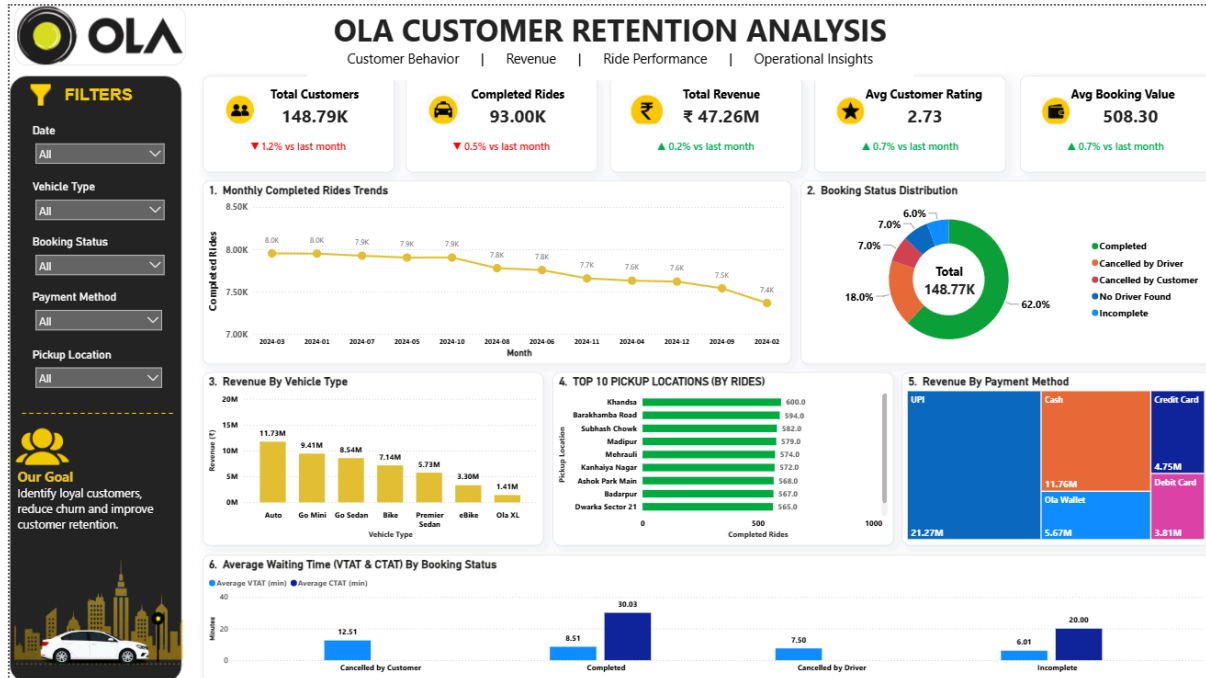
10. Dashboard Overview

The interactive dashboard includes:

- Revenue and trend analysis.
- Vehicle type and payment method distributions.
- Detailed cancellation and rating analysis.
- Real-time filters for executive decision-making.

The dashboard enables decision-makers to monitor operational performance and customer behavior in real time.

11. Dashboard Snapshot



12. Customer Behavior Analysis

Customer behavior analysis focuses on:

- Booking frequency
- Ride completion
- Preferred payment methods
- Ride distance
- Spending patterns
- Customer ratings

Key observations include:

High ride completion rates are directly linked to retention. Analysis shows a strong preference for digital payments, with UPI being the most frequently used method.

13. Cancellation Analysis

Ride cancellations are a primary pain point. The report captures reasons from both customers and drivers to identify operational bottlenecks and improve fulfillment rates.

Analysis includes:

- Customer cancellation reasons
- Driver cancellation reasons
- Cancellation frequency
- Cancellation trends
- Operational bottlenecks

14. Payment Analysis

Customers utilize several methods: UPI, Cash, Ola Wallet, and Credit/Debit cards. UPI's dominance suggests that digital-first incentives could further drive engagement.

Key Observation:

UPI is the most preferred payment method, followed by Cash and Ola Wallet. This indicates a strong customer preference for digital payments, which can be leveraged through cashback offers and loyalty programs.

15. Customer Satisfaction Analysis

Customer satisfaction is measured through:

- Customer Ratings
- Driver Ratings
- Ride Completion
- Ride Experience

With an average rating of 4.40/5, overall satisfaction is high. Maintaining these ratings is essential for ensuring repeat usage and long-term loyalty.

16. Key Business Insights

Based on the analysis:

- Over **93,000 rides** were successfully completed.
- Total revenue exceeded **₹51.8 million**.
- Average booking value is approximately **₹508**.
- Customer ratings remain consistently high, averaging **4.40/5**.
- UPI dominates as the preferred payment method.
- Ride cancellations remain a major factor affecting customer retention.
- Operational efficiency directly influences customer satisfaction.
- Faster ride fulfillment and fewer cancellations are likely to improve repeat usage.

17. Business Recommendations

1. **Reduce Driver Cancellations:** Implement performance monitoring and incentive programs.
2. **Improve Pickup Efficiency:** Optimize route planning to reduce wait times.
3. **Reward Loyal Customers:** Launch tier-based loyalty programs and referral credits.
4. **Promote Digital Payments:** Offer cashback for UPI and Ola Wallet transactions.
5. **Predict Customer Churn:** Use predictive analytics to engage at-risk customers.

18. Future Scope

The analysis can be expanded by:

- Customer churn prediction using Machine Learning
- Customer segmentation with clustering techniques
- Demand forecasting
- Dynamic pricing analysis
- Driver performance analytics
- Geographic heatmaps
- Real-time dashboard integration

- Sentiment analysis using customer feedback

19. Conclusion

The Ola Customer Retention Analysis demonstrates that data-driven insights into cancellations and payment behavior can significantly improve operational efficiency. By focusing on service quality and digital engagement, Ola can achieve sustainable long-term growth and enhanced customer loyalty.